

SHADOW 50M Instruct, asked twenty questions

A 44M-parameter model, 19.8 MB on disk, on a laptop CPU. Every answer below is the kernel's greedy output, unedited, one fresh process per question. The grey line under an answer is the working the model wrote before answering: a [calc] span is a circuit computing inside the forward pass, a [need] and [quote] pair is a record fetched from its memory on disk. Speed is the measured decode rate for that answer.

A word with it

1. Who are you?

I am SHADOW 50M, a small language model that runs locally on a CPU without internet.

24 tokens · 1,600 tokens/s · 41 MB

2. What is the capital of Japan?

From my knowledge: The capital of Japan is Tokyo.

13 tokens · 1,444 tokens/s · 41 MB

3. Who wrote Romeo and Juliet?

From my knowledge: William Shakespeare wrote Romeo and Juliet.

13 tokens · 1,625 tokens/s · 41 MB

4. Tell me a short joke about computers.

Why did the computer go to the doctor? Because it had a virus.

17 tokens · 1,700 tokens/s · 41 MB

5. Write a short poem about rain.

Rain falls on a soft, icy ceiling.

11 tokens · 1,571 tokens/s · 41 MB

Every circuit, as a word problem

6. I have 3 books and I bought 5 more. How many books do I have now?

You have 8 books.

[calc]3+5[eq]8[ecalc]

17 tokens · 1,546 tokens/s · 41 MB

7. A train had 48 passengers and 19 got off. How many passengers are left?

There are 29 passengers left.

[calc]48-19[eq]29[ecalc]
21 tokens · 1,750 tokens/s · 41 MB

8. There are 7 boxes with 12 eggs in each box. How many eggs is that in total?

84 eggs.

[calc]7*12[eq]84[ecalc]
17 tokens · 1,700 tokens/s · 41 MB

9. I split 96 apples equally into 8 bags. How many apples are in each bag?

12

[calc]96//8[eq]12[ecalc]
15 tokens · 1,500 tokens/s · 41 MB

10. My bill is 240 dollars. What is 15 percent of that?

36 dollars.

[calc]pct:15% of 240[eq]36[ecalc]
24 tokens · 1,600 tokens/s · 41 MB

11. What date is 45 days after December 20, 2026?

February 3, 2027.

[calc]cal:Dec20+45d[eq]February 3[ecalc]
27 tokens · 1,588 tokens/s · 41 MB

12. My exam is on March 14, 2026. What day of the week is that?

Saturday.

[calc]dow:Mar14[eq]Saturday[ecalc]
16 tokens · 1,600 tokens/s · 41 MB

13. My suitcase weighs 23 kg. What is that in pounds?

50.7 pounds.

[calc]unit:23kg>lb[eq]50.7[ecalc]
24 tokens · 1,200 tokens/s · 41 MB

14. How many times does the letter s appear in the word mississippi?

4.

[calc]count:s in mississippi[eq]4[ecalc]
15 tokens · 1,667 tokens/s · 41 MB

15. Sort these from smallest to largest: 84, 19, 507, 3, 261.

3,19,84,261,507

[calc]cmp:sort 84,19,507,3,261[eq]3,19,84,261,507[ecalc]

16. Take 200, add 50, subtract 30, multiply by 3, then add 9. What do you get?

669

[calc]200+50[eq]250[ecalc]

[calc]250-30[eq]220[ecalc]

[calc]220*3[eq]660[ecalc]

[calc]660+9[eq]669[ecalc]

56 tokens · 2,000 tokens/s · 41 MB

From its memory on disk (eight records, three of them about patients)

Patient P-204 condition: asthma.
Patient P-204 allergy: penicillin.
Patient R-118 condition: hypertension.
Patient R-118 allergy: none recorded.
Sample K-77 weight in kg: 12.
Sample K-77 temperature in C: 21.
Parcel M-3310 weight in kg: 40.
Vault-418 count: 3271.

17. What is the condition of Patient P-204?

The condition of Patient P-204 is asthma.

[need]condition of Patient P-204.

[quote]pos=1: Patient P-204 condition: asthma.

37 tokens · 1,423 tokens/s · 42 MB

18. What is the allergy of Patient P-204?

The allergy of Patient P-204 is penicillin.

[need]allergen of Patient P-204.

[quote]pos=2: Patient P-204 allergy: penicillin.

39 tokens · 1,182 tokens/s · 42 MB

19. Look up the weight in kg of Sample K-77 and give it in pounds.

26.5 pounds

[need]weight in kg of Sample K-77.

[quote]pos=5: Sample K-77 weight in kg: 12.

[calc]unit:12kg>lb[eq]26.5[ecalc]

51 tokens · 1,159 tokens/s · 42 MB

20. What is the condition of Patient Z-999?

There is no record of Patient Z-999.

[need]condition of Patient Z-999.

searched the records: no record of Patient Z-999 found.

36 tokens · 1,091 tokens/s · 41 MB

Chosen from what it does well; its poems are one line long, seven-digit operands sometimes arrive in a span wrong, and the full list of what it gets wrong is in the repository. github.com/QLNI/SHADOW-50M-Instruct · huggingface.co/QLNI/shadow-50m-instruct · MIT · © QLNI 2026